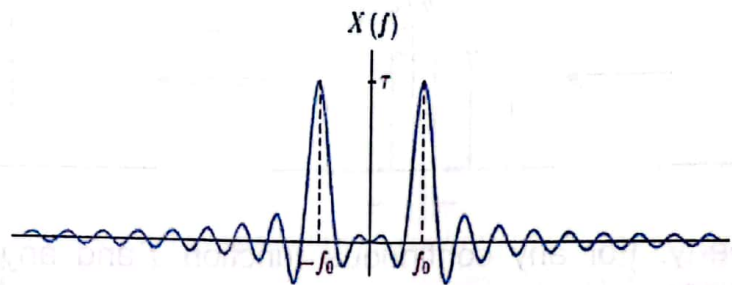


AL-Ayen Iraqi University

Engineering Collage / AI Engineering Department

Signals and Systems

Lecture 2 : Basic continuous-time signals



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المرحلة التالية
إشارات ونظم
المحاضرة الثانية

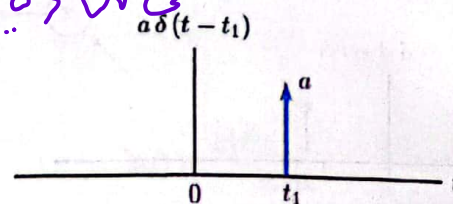
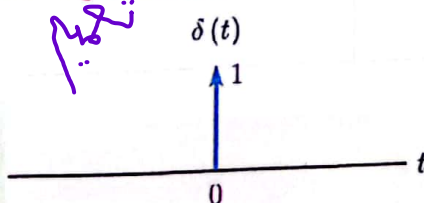
Amjed Jumaah

Unit-impulse function

- The unit-impulse function (Dirac delta function or delta function), denoted δ , is defined by:

$$\delta(t) = \begin{cases} 0, & \text{if } t \neq 0 \\ \text{undefined,} & \text{if } t = 0 \end{cases} \quad \text{and} \quad \int_{-\infty}^{\infty} \delta(t) dt = 1$$

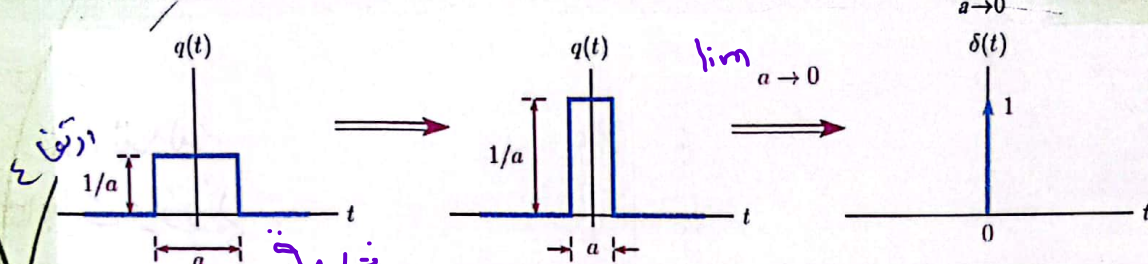
- Technically, δ is not a function in the ordinary sense. Rather, it is what is known as a generalized function.



Define $q(t) = \begin{cases} 1/a, & |t| < a/2 \\ 0, & |t| > a/2 \end{cases}$

Clearly, for any choice of a , $\int_{-\infty}^{\infty} q(t) dt = 1$

The function δ can be obtained as the following limit: $\delta(t) = \lim_{a \rightarrow 0} q(t)$



Sampling property. For any continuous function f and any real constant t_1 , $f(t)\delta(t-t_1) = f(t_1)\delta(t-t_1)$.

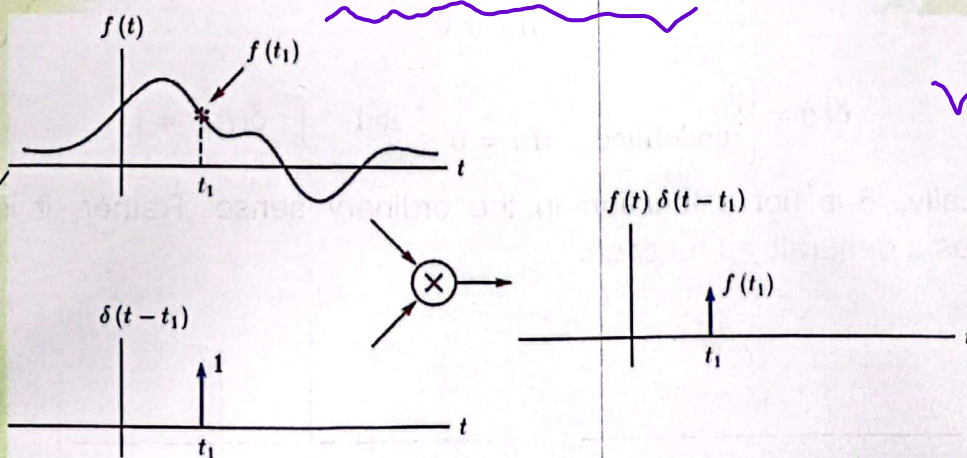
النتيجة دالو Unit Impuls

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1

Sifting property. For any continuous function f and any real constant t_1 :-

$$\int_{-\infty}^{\infty} f(t)\delta(t-t_1)dt = f(t_1)$$

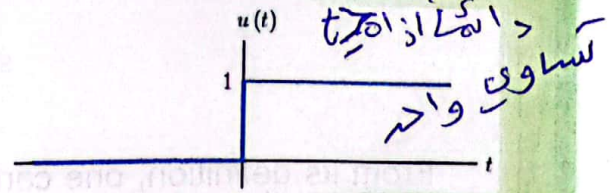


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Unit-Step Function

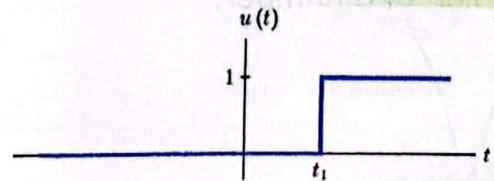
- The **unit-step function** (also known as the **Heaviside function**), denoted u , is defined as:

$$u(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$$



- A time **shifted version** of the unit-step function:

$$u(t - t_1) = \begin{cases} 1, & t \geq t_1 \\ 0, & t < t_1 \end{cases}$$



- Signals begin at $t = 0$ (**causal signals**) can be described in terms of $u(t)$.

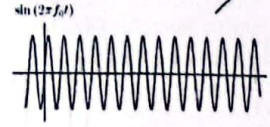
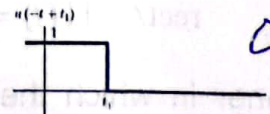
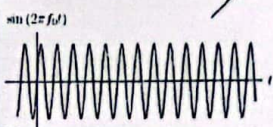
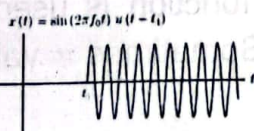
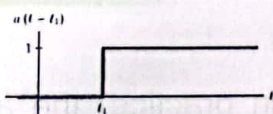
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2

- Using the **unit-step function** to **turn a signal on/off** at a specified time instant: $\Rightarrow$

$$x(t)u(t - t_1) = \begin{cases} \sin(2\pi f_0 t), & t \geq t_1 \\ 0, & t < t_1 \end{cases}$$

$$x(t)u(-t + t_1) = \begin{cases} \sin(2\pi f_0 t), & t \leq t_1 \\ 0, & t > t_1 \end{cases}$$



تغییر بعد از این زمان
بالا نیاورد
اطلاعات
منتقله

- The **Relationship** between the **unit-step function** and the **unit-impulse function**:

$$\delta(t) = \frac{du(t)}{dt}$$

$$u(t) = \int_{-\infty}^t \delta(\tau) d\tau$$

Unit Impulse
تغییر از زمان

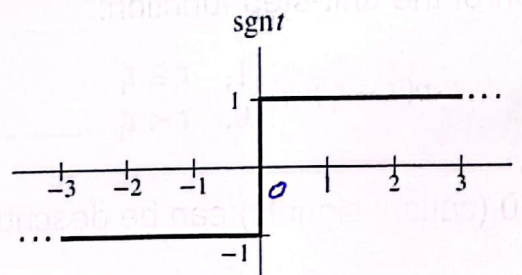
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Signum Function

- The **signum function**, denoted sgn , is defined as:

$$\text{sgn} t = \begin{cases} 1 & \text{if } t > 0 \\ 0 & \text{if } t = 0 \\ -1 & \text{if } t < 0 \end{cases}$$

From its definition, one can see that the signum function simply computes the **sign** of a number.



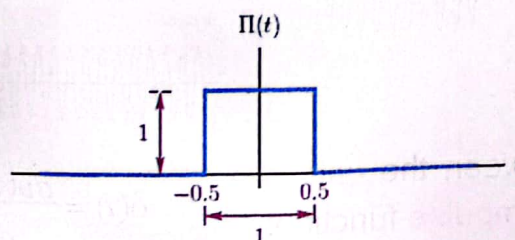
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Unit-pulse function

- The **unit-pulse function** ^{دالة النبضة}, denoted rect , is given by: (also called the unit-rectangular pulse function),

$$\text{rect} t = \Pi(t) = \begin{cases} 1, & \text{if } -\frac{1}{2} < t < \frac{1}{2} \\ 0, & \text{otherwise} \end{cases}$$

- Due to the manner in which the rect function is used in practice, the actual **value of $\text{rect} t$ at $t = \pm \frac{1}{2}$** is unimportant. Sometimes $\neq$ values are used.

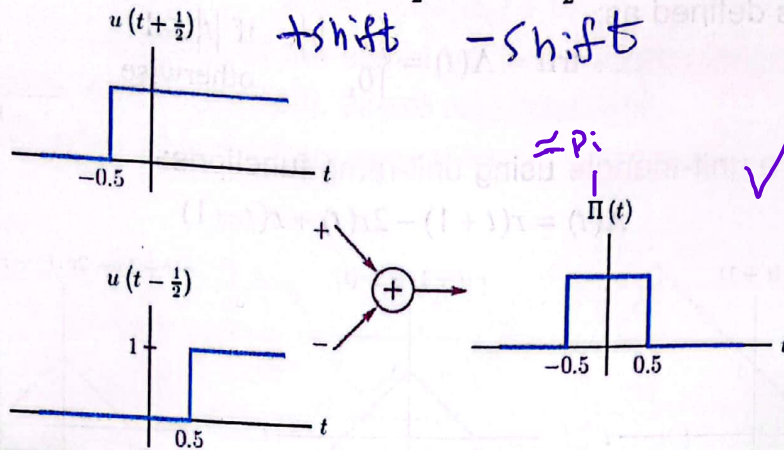


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3

- Constructing a **unit-pulse** function from **unit-step** functions:

$$\Pi(t) = u(t + \frac{1}{2}) - u(t - \frac{1}{2})$$



Unit-Ramp Function.

- The **unit-ramp** function, denoted r , is defined as:

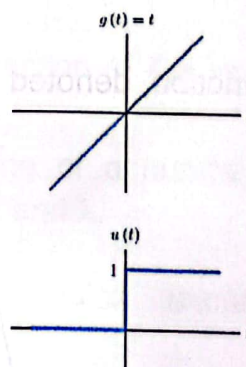
$$r(t) = \begin{cases} t, & \text{if } t \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

or, equivalently: $r(t) = tu(t)$.

- Constructing a **unit-ramp** function from a **unit-step**:

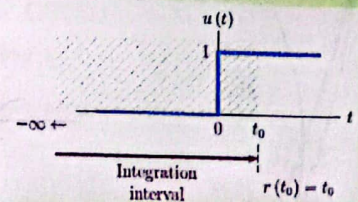
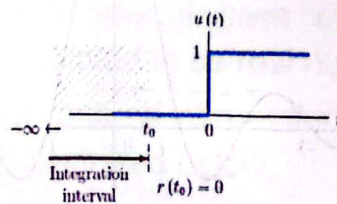
$$r(t) = \int_{-\infty}^t u(\tau) d\tau$$

ramp



amplitude if $t \geq 0$

linear

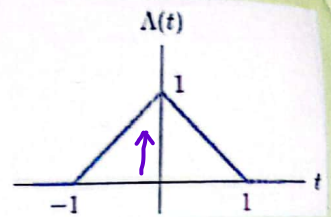


Unit Triangular Function

- The **unit triangular function** (unit-triangular pulse function),

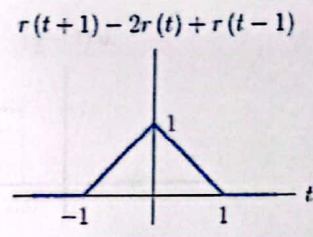
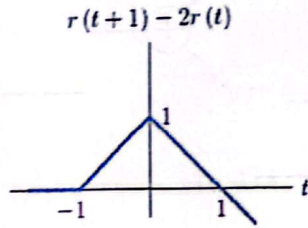
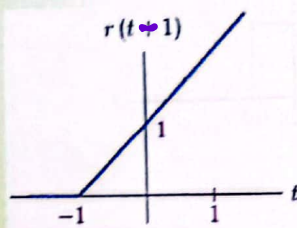
denoted tri , is defined as:

$$\text{tri } t = \Lambda(t) = \begin{cases} 1 - |t|, & \text{if } |t| \leq 1 \\ 0, & \text{otherwise} \end{cases}$$



- Constructing a **unit-triangle** using **unit-ramp** functions:-

$$\Lambda(t) = r(t+1) - 2r(t) + r(t-1)$$

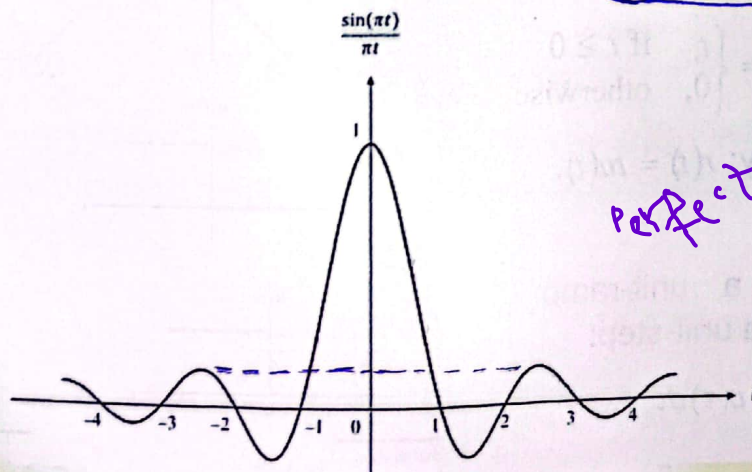


تكون الإشارة المثلثية

Cardinal Sine Function

- The **cardinal sine function**, denoted sinc , is given by

$$\text{sinc } t = \frac{\sin(\pi t)}{\pi t}$$



Sinusoidal Signal

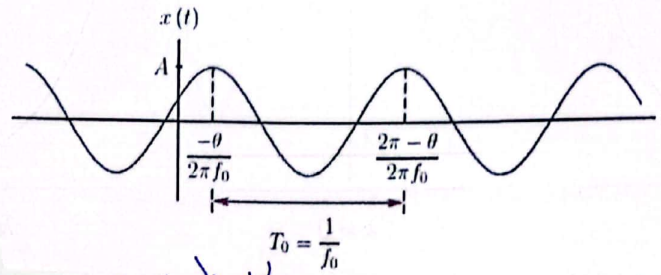
الانماذج الجيبية

- A **real sinusoidal function** is a function of the form:

$$x(t) = A \cos(\omega_0 t + \theta)$$

where A is the **amplitude** of the signal, ω_0 is the **radian frequency** (rad/s), and θ is the initial phase angle (rad), all are **real** constants.

$\omega_0 = 2\pi f_0$ where f_0 is the **frequency** (Hz), $T_0 = 1/f_0$ is the **period** (s).

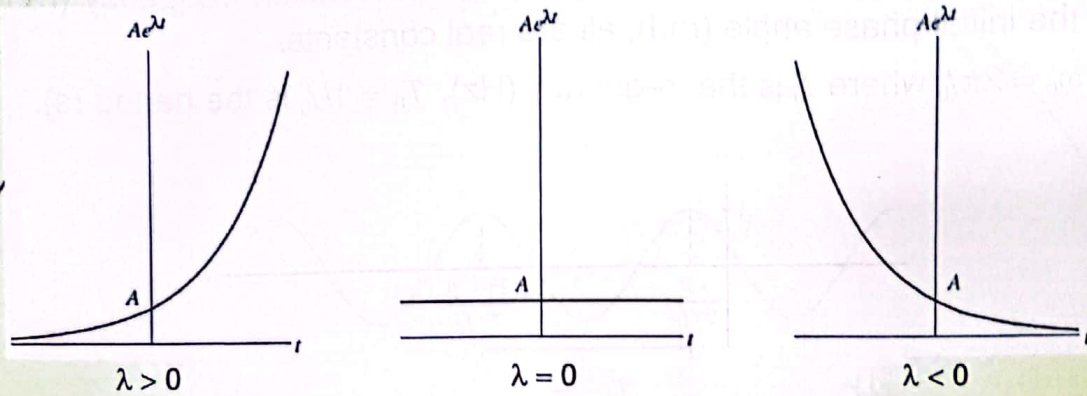


Period $\leftarrow T_0 = \frac{1}{f_0}$

Complex Exponential Function

- A **complex exponential** function is a function of the form $x(t) = Ae^{\lambda t}$, where A and λ are complex **constants**.
- A complex exponential can exhibit one of a number of **distinct modes of behavior**, depending on the values of A and λ .
- For example, as special cases, complex exponentials include real exponentials and complex sinusoids.
- A **real exponential** function is a special case of a complex exponential $x(t) = Ae^{\lambda t}$, where A and λ are restricted to be **real** numbers.
- A real exponential can exhibit one of **three distinct modes** of behavior, depending on the value of λ , as illustrated below.

- If $\lambda > 0$, $x(t)$ **increases** exponentially as t increases (growing exponential).
- If $\lambda < 0$, $x(t)$ **decreases** exponentially as t increases (decaying exponential).
- If $\lambda = 0$, $x(t)$ simply equals the **constant** A .

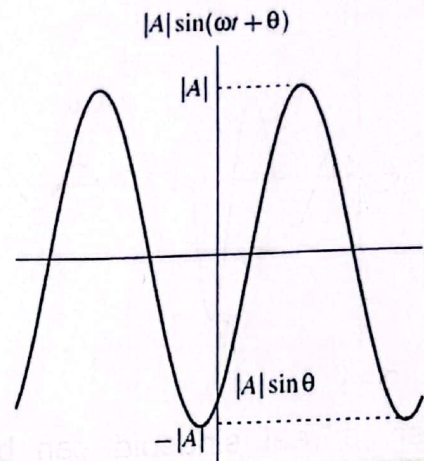
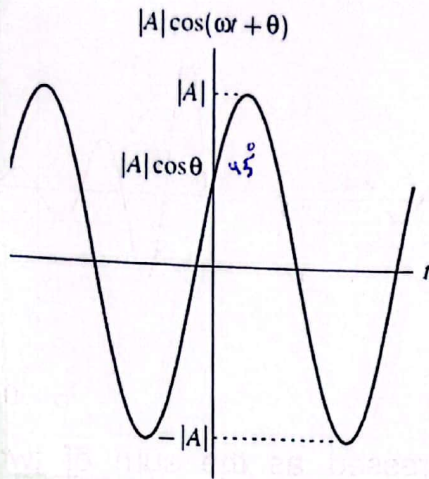


Exponential دالة مرتفعة مع الزمن او متناقصة مع الزمن
 وهي دالة تصل الى صفر عالية جدا وتقترب من الصفر جدا

Complex Sinusoidal Function

تكون من جزئين حقيقي وتخيالي

- A **complex sinusoidal function** is a special case of a complex exponential $x(t) = Ae^{j\omega t}$, where A is **complex** and λ is **purely imaginary** (i.e., $\text{Re}\{\lambda\} = 0$).
- That is, a **complex sinusoidal function** is a function of the form $x(t) = Ae^{j\omega t}$, where A is **complex** and ω is **real**.
- By expressing A in polar form as $A = |A|e^{j\theta}$ (where θ is real) and using Euler's relation, we can rewrite $x(t)$ as: $x(t) = \underbrace{|A| \cos(\omega t + \theta)}_{\text{Re}\{x(t)\}} + \underbrace{j|A| \sin(\omega t + \theta)}_{\text{Im}\{x(t)\}}$
- Thus, $\text{Re}\{x\}$ and $\text{Im}\{x\}$ are the same except for a time shift.
- Also, x is periodic with **fundamental period** $T = 2\pi/|\omega|$ and **fundamental frequency** $|\omega|$.



- In the most general case of a complex exponential function $x(t) = Ae^{\lambda t}$, A and λ are both **complex**.

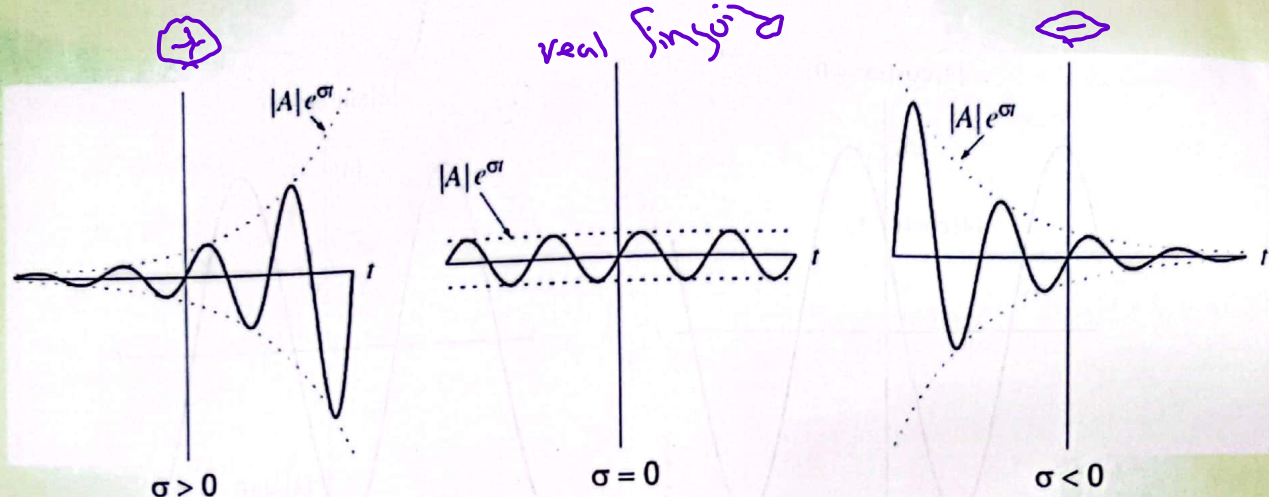
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لغات من السطح

- Letting $A = |A|e^{j\theta}$ and $\lambda = \sigma + j\omega$ (where θ , σ , and ω are real), and using Euler's relation, we can rewrite $x(t)$ as: $x(t) = \underbrace{|A|e^{\sigma t}\cos(\omega t + \theta)}_{\text{Re}\{x(t)\}} + j \underbrace{|A|e^{\sigma t}\sin(\omega t + \theta)}_{\text{Im}\{x(t)\}}$
- Three distinct modes** depending on the value of σ :
 - If $\sigma = 0$, $\text{Re}\{x\}$ and $\text{Im}\{x\}$ are **real sinusoids**.
 - If $\sigma > 0$, $\text{Re}\{x\}$ and $\text{Im}\{x\}$ are each the **product of a real sinusoid and a growing real exponential**.
 - If $\sigma < 0$, $\text{Re}\{x\}$ and $\text{Im}\{x\}$ are each the **product of a real sinusoid and a decaying real exponential**.
- From Euler's relation, a complex sinusoid can be expressed as the sum of two real sinusoids as:

حسب النماذج

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- Moreover, a real sinusoid can be expressed as the sum of two complex sinusoids using the identities:

$$A \cos(\omega t + \theta) = A [e^{j(\omega t + \theta)} + e^{-j(\omega t + \theta)}] \quad \text{and} \quad A \sin(\omega t + \theta) = A [e^{j(\omega t + \theta)} - e^{-j(\omega t + \theta)}]$$

Assignments

- 1- Use the Sifting Property to evaluate:

$$\int_{-\infty}^{\infty} f(t) \delta(t - 3) dt$$

where $f(t) = t^2 + 2t$.

- 2- Find the period T and frequency f of the sinusoidal signal:

$$x(t) = 5 \cos \left(20\pi t + \frac{\pi}{4} \right)$$

THANK YOU